AMRITA VISHWA VIDYAPEETHAM CHENNAI CAMPUS

Amrita School of Computing

Coding Practice

Activity Sheets

(For the Batch 2023-2027)

Name:

Roll No.:

Department:

Instructions

The Mock Test syllabus for CSE CT | CYS | AIE | RAI program is enclosed in the attached document for your reference.

Materials for the Mock Test [Must-DO Coding Questions] is provided in a link.

It is mandatory to upload the completed Activity sheets before the deadline in the MS-Form link shared by your class Advisor.

Activity Sheet: 01 - Coding Questions from Sorting and Searching – Part 1 [Easy + Medium Questions]

Each Activity Sheets contains total of 20 Questions.

Platform: Any Online Compiler of your choice – GDB Compiler

Language: C | C++ | JAVA | PYTHON

The Activity Sheet will have the coding Questions and the Test Case. Once executed in Online Compiler, take screen shot of the code and the output and paste it in the space provided in the activity sheet.

The Final Activity Sheet should be in PDF Format and the file name should be your roll number.

Course Instructors: CARE TEAM
Dr.R.Annamalai Assistant Professor (Sl.Grade) |CSE-AIE
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1. Maximize array sum after K negations using Sorting
Given an array of size n and an integer k. We must modify array k number of times. In each modification, we can replace any array element arr[i] by -arr[i]. The task is to perform this operation in such a way that after k operations, the sum of the array is maximum.

Examples :

Input : arr[] = [-2, 0, 5, -1, 2], k = 4
Output: 10

Explanation:
1. Replace (-2) by -(-2), array becomes [2, 0, 5, -1, 2]
2. Replace (-1) by -(-1), array becomes [2, 0, 5, 1, 2]
3. Replace (0) by -(0), array becomes [2, 0, 5, 1, 2]
4. Replace (0) by -(0), array becomes [2, 0, 5, 1, 2]

Input : arr[] = [9, 8, 8, 5], k = 3
Output: 20

Explanation: Negate 5 three times. Array will become [9, 8, 8, -5].

Code:

Output:

2. Sum of minimum absolute differences in an array

Given an array of n distinct integers. The task is to find the sum of minimum absolute difference of each array element. For an element arr[i] present at index i in the array, its minimum absolute difference is calculated as: Min absolute difference (arr[i]) = min(abs(arr[i] - arr[j])), where 0 <= j < n and j != i and abs is the absolute value.

Examples:
Input : arr = [4, 1, 5]
Output : 5
Explanation: Sum of minimum absolute differences is |4-5| + |1-4| + |5-4| = 1 + 3 + 1 = 5

Input : arr = [5, 10, 1, 4, 8, 7]
Output : 9
Explanation: Sum of minimum absolute differences is
|5-4| + |10-8| + |1-4| + |4-5| + |8-7| + |7-8| = 1 + 2 + 3 + 1 + 1 + 1 = 9

Input : arr = [12, 10, 15, 22, 21, 20, 1, 8, 9]
Output : 18

Code:

Output

3. Given an array arr[] and an integer sum, check if there is a triplet in the array which sums up to the given target sum.

Examples:

Input: arr[] = [1, 4, 45, 6, 10, 8], target = 13
Output: true
Explanation: The triplet [1, 4, 8] sums up to 13

Input: arr[] = [1, 2, 4, 3, 6, 7], target = 10
Output: true
Explanation: The triplets [1, 3, 6] and [1, 2, 7] both sum to 10.

Input: arr[] = [40, 20, 10, 3, 6, 7], sum = 24
Output: false
Explanation: No triplet in the array sums to 24.

Code:

Output

4. Given an array citations[] of size n such that citations[i] is the number of citations a researcher received for ith paper, find the H-index.

H-index(H) is the largest value such that the researcher has published at least H papers that have been cited at least H times.

'H' stands for Hirsch index as it was proposed by the J.E. Hirsch in 2005. The H-index is defined as the author-level metric that attempts to measure both the productivity and the citation impact of the publication of the scientist or the scholar.

Examples:

Input: citations[] = [5, 0, 2, 0, 2]
Output: 2
Explanation: We can see that there are at least 2 papers whose citation count is 2 or more. In this case, the papers with 5, 2, and 2 citations qualify, and since at least 2 such papers exist, the H-index is 2.

Input: citations[] = [6, 0, 3, 5, 3]
Output: 3
Explanation: Here, there are at least 3 papers that have been cited 3 or more times. The papers with 6, 5, 3, and 3 citations meet this condition. Since at least 3 such papers exist, the H-index is 3.

Code:

Output

5. Given an integer array arr[] and an integer target, find the sum of triplets such that the sum is closest to target.

Note: If there are multiple sums closest to target, print the maximum one.

Examples:

Input: arr[] = [-1, 2, 2, 4], target = 4
Output: 5
Explanation: All possible triplets
[-1, 2, 2], sum = (-1) + 2 + 2 = 3
[-1, 2, 4], sum = (-1) + 2 + 4 = 5
[-1, 2, 4], sum = (-1) + 2 + 4 = 5
[2, 2, 4], sum = 2 + 2 + 4 = 8
Triplet [-1, 2, 2], [-1, 2, 4] and [-1, 2, 4] have sum closest to target, so return the maximum one, that is 5.

Input: arr[] = [1, 10, 4, 5], target = 10
Output: 10
Explanation: All possible triplets
[1, 10, 4], sum = (1 + 10 + 4) = 15
[1, 10, 5], sum = (1 + 10 + 5) = 16
[1, 4, 5], sum = (1 + 4 + 5) = 10
[10, 4, 5], sum = (10 + 4 + 5) = 19
Triplet [1, 4, 5] has sum = 10 which is closest to target.

Code:

Output

6. Given two sorted arrays a[] and b[] of size n and m respectively, merge both the arrays and rearrange the elements such that the smallest n elements are in a[] and the remaining m elements are in b[]. All elements in a[] and b[] should be in sorted order.

Examples:

Input: a[] = [2, 4, 7, 10], b[] = [2, 3]
Output: a[] = [2, 2, 3, 4], b[] = [7, 10]
Explanation: Combined sorted array = [2, 2, 3, 4, 7, 10], array a[] contains smallest 4 elements: 2, 2, 3 and 4, and array b[] contains remaining 2 elements: 7, 10.

Input: a[] = [1, 5, 9, 10, 15, 20], b[] = [2, 3, 8, 13]
Output: a[] = [1, 2, 3, 5, 8, 9], b[] = [10, 13, 15, 20]
Explanation: Combined sorted array = [1, 2, 3, 5, 8, 9, 10, 13, 15, 20], array a[] contains smallest 6 elements: 1, 2, 3, 5, 8 and 9, and array b[] contains remaining 4 elements: 10, 13, 15, 20.

Input: a[] = [0, 1], b[] = [2, 3]
Output: a[] = [0, 1], b[] = [2, 3]
Explanation: Combined sorted array = [0, 1, 2, 3], array a[] contains smallest 2 elements: 0 and 1, and array b[] contains remaining 2 elements: 2 and 3.

Code:

Output

7. Given an unsorted array arr[] of distinct integers, construct another array countSmaller[] such that countSmaller[i] contains the count of smaller elements on the right side of each element arr[i] in the array.
Examples:

Input: arr[] = [12, 1, 2, 3, 0, 11, 4]
Output: [6, 1, 1, 1, 0, 1, 0]

Input: arr[] = [5, 4, 3, 2, 1]
Output: [4, 3, 2, 1, 0]

Input: arr[] = [1, 2, 3, 4, 5]
Output: [0, 0, 0, 0, 0]

Code:

Output

8. Given an array of positive numbers, the task is to find the smallest positive integer value that cannot be represented as the sum of elements of any subset of a given set.

Examples:

Input: arr[] = {1, 10, 3, 11, 6, 15};
Output: 2

Input: arr[] = {1, 1, 1, 1};
Output: 5

Input: arr[] = {1, 1, 3, 4};
Output: 10

Code:

Output

9. Three arrays of same size are given. Find a triplet such that maximum - minimum in that triplet is minimum of all the triplets. A triplet should be selected in a way such that it should have one number from each of the three given arrays.

If there are 2 or more smallest difference triplets, then the one with the smallest sum of its elements should be displayed.

Examples :

Input : arr1[] = {5, 2, 8}
arr2[] = {10, 7, 12}
arr3[] = {9, 14, 6}
Output : 7, 6, 5

Input : arr1[] = {15, 12, 18, 9}
arr2[] = {10, 17, 13, 8}
arr3[] = {14, 16, 11, 5}
Output : 11, 10, 9

Code:

Output

10. Given two arrays a[] and b[] of equal size and an integer k, Determine whether there exists any permutation of the arrays such that for every index i, the condition a[i] + b[i] ≥ k holds. Return true if such a permutation exists, otherwise return false.
Examples :

Input: a[] = [2, 1, 3], b[] = [7, 8, 9], k = 10
Output: true
Explanation: Rearranging gives a = [1, 2, 3] and b = [9, 8, 7]. Each pair sum (10, 10, 10) is ≥ k = 10

Input: a[] = [1, 2, 2, 1], b[] = [3, 3, 3, 4], k = 5
Output: false
Explanation: Even after rearranging both arrays, at least one pair sum becomes less than k = 5.
Hence, no valid permutation exists, and the answer is false.

Code:

Output

11. Minimum Jumps to Reach End

Given arr[] where arr[i] is the max steps you can jump from index i, find the minimum number of jumps to reach the last index from index 0. Return -1 if impossible.

Constraints:
1 <= n <= 10^4 | 0 <= arr[i] <= 10^4

Input
arr=[2,3,1,1,4]
Output
2 (0->1->4)

Input
arr=[2,3,0,1,4]
Output
2 (0->1->4)

Input
arr=[0,2,3]
Output
-1

Code:

Output

12. Three Closest Elements to a Target

Given a sorted array arr[] and a target x, find the three elements closest to x. Return them in sorted order. In case of tie in distance, prefer the smaller element.

Constraints: 3 <= n <= 10^5 | -10^9 <= arr[i], x <= 10^9 | All elements distinct

Input:
arr=[1,2,3,4,5,6,7,8,9], x=5
Output:
[4,5,6]
Input:
arr=[10,20,30,40,50], x=25
Output:
[20,30,40]
Input:
arr=[-10,-5,0,3,7,15], x=6
Output:
[3,7,15]

Code:

Output

13. Given two sorted arrays arr1[] and arr2[] of sizes m and n respectively, find the k-th element (1-indexed) in their combined sorted order. Expected time: O(log(min(m,n))).

Constraints:

$1 \leq m, n \leq 10^6$ | $1 \leq k \leq m+n$ | $0 \leq arr1[i], arr2[i] \leq 10^6$

Example Test Cases:

Input
arr1=[2,3,6,7,9], arr2=[1,4,8,10], k=5
Output
6

Input
arr1=[1,2,3], arr2=[4,5,6], k=4
Output
4
Input
arr1=[0,0,0], arr2=[0,0,0], k=3
Output
0

Code:

Output

14. Given n stalls at positions arr[] and c cows, place the cows such that the minimum distance between any two cows is maximized. Return that maximum minimum distance.

Constraints:
$2 \leq n \leq 10^5$ | $2 \leq c \leq n$ | $0 \leq arr[i] \leq 10^9$

Input:
arr=[0,3,4,7,10,9], c=4
Output:
3

Input:
arr=[1,2,4,8,9], c=3
Output:
3

Code:

Output

15. Given a sorted list of numbers in which all elements appear twice except one element that appears only once, find the number that appears only once.

Constraints
1 <= T <= 100
1 <= Size of array <= 20000
1 <= Elements in array <= 106

Example:
1, 1, 2, 3, 3, 4, 4
Here, ‘2’ appears once and all other elements appear twice.

findNonRepeatingElement([1, 1, 2, 3, 3, 4, 4]) => 2
Expected Time Complexity: O(log n)

Testing
Input Format
The first line contains 'T', denoting the no. of test cases.

Each test case consists of two lines. The first contains a number 'n' denoting the number of elements. The second line has ‘n’ space-separated numbers denoting the elements.

Output Format
For each test case, a line containing the non-repeating number.

Sample Input
3
1
3
3
1 2 2
5
3 3 4 4 5
Expected Output
3
1
5

Code:

Output

16. You are given a list of unique integers which are sorted but rotated at some pivot. You are also given a target value and you have to find its index in the list. If it is not present in the list, return -1.

Constraints
1 <= T <= 100
1 <= n <= 104
1 <= array elements <= 106
1 <= target <= 106

Example:
List: [4, 5, 6, 7, 1, 2, 3]
Target value: 6
Resultant index: 2

Testing
Input Format
The first line contains 'T', denoting the number of test cases.

Each test contains 3 lines:

a number 'n', denoting the number of elements.
n space-separated numbers denoting the array elements.
the target value.
Output Format
T lines, each containing a number denoting the index of the target value. -1 if the target value is not present.

Sample Input
4
7
4 5 6 7 0 1 2
4
4
3 4 1 2
5
5
5 1 2 3 4
2
4
5 6 3 4
4
Expected Output
0
-1
2
3

Code:

Output

17. Wave Array

Given an unsorted array arr[], rearrange it in-place into a wave pattern:
arr[0]>=arr[1]<=arr[2]>=arr[3]<=arr[4] and so on. Print any valid wave arrangement.

Note: Sort first (O(n log n)), then swap every adjacent pair at even indices (O(n)). This guarantees the wave property.

Constraints:
1 <= n <= 10^5 | 0 <= arr[i] <= 10^9

Input
arr=[10,5,6,3,2,20,100,80]
Output
[10,5,6,2,20,3,100,80] (one valid answer)
Input
arr=[1,2,3,4]
[2,1,4,3]
Input
arr=[5,1,3]
[5,1,3] (one valid answer)

Code:

Output

18. Find Pair with Maximum Product
Given an integer array arr[] (may contain negatives), find the pair of elements whose product is maximum. Return the product.
Note: The answer is either the product of the two largest positive numbers OR the two most-negative numbers.

Constraints:
2 <= n <= 10^5 | -10^4 <= arr[i] <= 10^4

Input
arr=[1,4,3,6,7,0]
Output
42 (6x7)

Input
arr=[-10,-3,5,6,-2]
Output
30 (-10x-3)

Input
arr=[-1,-2,-3,-4]
Output
12 (-3x-4)

Code:

Output

19. Given a sorted array containing distinct integers and a number 'key', find the index of the key in the array. If the key is not present, return the index at which it would be inserted considering that we need to maintain the sort order.

Constraints
1 <= T <= 100
1 <= n <= 104
-106 <= Ai <= 106
-106 <= key <= 106

Examples
Array: [1, 2, 3, 4, 5]
Number: 3
Answer: 2

Array: [1, 2, 3, 5]
Number: 4
Answer: 3

Array: [1, 2, 3, 4, 5]
Number: -100
Answer: 0

Array: [1, 2, 3, 4, 5]
Number: 6
Answer: 5

Testing
Input Format
First-line contains an integer 'T' denoting the number of test cases.
For each test case the input has two lines:
Two space-separated integers 'n' and 'key' denoting the length of the array and the number to be searched respectively.
n space-separated integers denoting the elements of the array.
Output Format
T lines each contain the index of the first occurrence of the key in the array or the index to insert the number (if not present) for each test case.

Sample Input
4
5 3
1 2 3 4 5
4 4
1 2 3 5
5 -100
1 2 3 4 5
5 6
1 2 3 4 5
Expected Output
2
3
0
5

Code:

Output

20. Given a sorted array and a number key, find the smallest array element which is greater than the key.

If the key is greater than or equal to the largest element then return the key itself.

Constraints
1 <= T <= 100
1 <= n <= 104
-106 <= Ai <= 106
-106 <= key <= 106

Examples
Array: [1, 2, 3, 3, 4, 4, 8, 10]
Number: 4
Answer: 8

Array: [1, 2, 3, 3, 3, 4, 4, 5]
Number: 5
Answer: 5 (Largest Element)

Array: [1, 2, 3, 3, 3, 4, 4, 5]
Number: -5
Answer: 1

Testing
Input Format
First-line contains an integer ‘T’ denoting the number of test cases.
For each test case the input has two lines:
Two space-separated integers ‘n’ and 'key' denoting the length of the array and the number to be searched respectively.
n space-separated integers denoting the elements of the array.
Output Format
T lines each contain the next greater element for each test case.

Sample Input
2
8 4
1 2 3 3 4 4 8 10
8 5
1 2 3 3 3 4 4 5
Expected Output
8
5

Code:

Output